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PRESIDENCY UNIVERSITY

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School of Computer Science Engineering & Information Science

A Report on
RFID Technology

Design and Implementation of a Smart Library Management System

Course Title: Privacy and security in IOT

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1 Introduction

Radio Frequency Identification (RFID) technology offers significant advantages over tra ditional barcode systems, particularly in environments requiring rapid, non-line-of-sight tracking. The implementation of RFID in library systems represents a crucial step toward digitizing inventory management and enhancing user experience.

1.1 Objective of the Report

The objective is to design a prototype for a **Smart Library Management System (LMS)** that automates the check-in/check-out process, tracks book inventory, and provides basic anti-theft capabilities using the principles of RFID.

1.2 Fundamental Physics

The system operates at 13.56 MHz (HF band), relying on **inductive coupling** where the reader's magnetic field energizes the tag's coil. The power transmission occurs over a short range, typically less than one meter, defined by the coupling efficiency.

2 RFID System Architecture

The LMS architecture must support both instantaneous transaction processing (check in/out) and long-term data storage.

2.1 Core Components

1. **Book Tag:** HF RFID tag adhered inside the book cover. Stores the ISBN/Asset ID.

2. **User Tag:** HF RFID tag in the student/librarian ID card.

3. **RFID Reader/Writer:** Used at the circulation desk to read/update the status.

4. **Database System:** The central server that links the UID to the book title, student name, and loan status.

2.2 Architectural Data Flow

The diagram illustrates the flow from the physical transaction (tag scan) to the centralized inventory system.

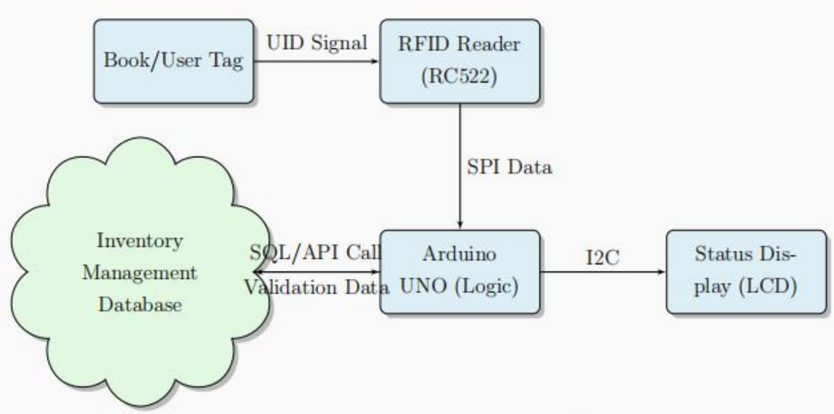

Figure 1: System Architecture of Smart Library Management

3 Smart Library Management System

3.1 Problem Statement

Traditional library operations face challenges in:

• **Manual Entry:** Slow and error-prone process during checkout.

• **Inventory Discrepancy:** Difficulty performing quick shelf-reading and stock ver ification.

• **Security Lapses:** Easy to conceal books from older magnetic security strips.

3.2 Check-in / Check-out Logic

The process relies on a simple state transition:

1. Scan Book Tag: UID is read.

2. Database Check: If the UID is marked 'Checked Out', the system changes the status to 'Available' (Check-in).

3. Database Check: If the UID is marked 'Available', the system logs the user and changes the status to 'Checked Out' (Check-out).

4 Working Model Implementation

4.1 Components Used

The prototype utilizes low-cost components to simulate the high-frequency library envi ronment.

Table 1: List of Hardware Components for LMS Prototype

Component	**Model**	**Purpose**
Microcontroller	Arduino UNO	Manages logic and communication.
Reader Module	RC522	Reads 13.56MHz Book/User Tags.
Tags	Mifare 1K (Multiple)	Simulated Books and User Cards.
Output	Status LED / Buzzer	Provides transaction feedback.

5 Conclusion and Future Scope

This report successfully demonstrated the feasibility of implementing a **Smart Library Management System** using the RC522 RFID module. This application highlights the power of RFID in **data management and non-line-of-sight inventory control**, address ing critical issues faced by traditional libraries.

5.1 Future Enhancements

For a professional deployment, the system must be upgraded to UHF RFID for simulta neous scanning of multiple books (anti-collision) and long-range inventory counting. The Arduino component must be replaced with an **ESP32** or **Raspberry Pi** to enable direct API communication with the central Presidency University library database.